

ILN Rubric: IR (20 points)

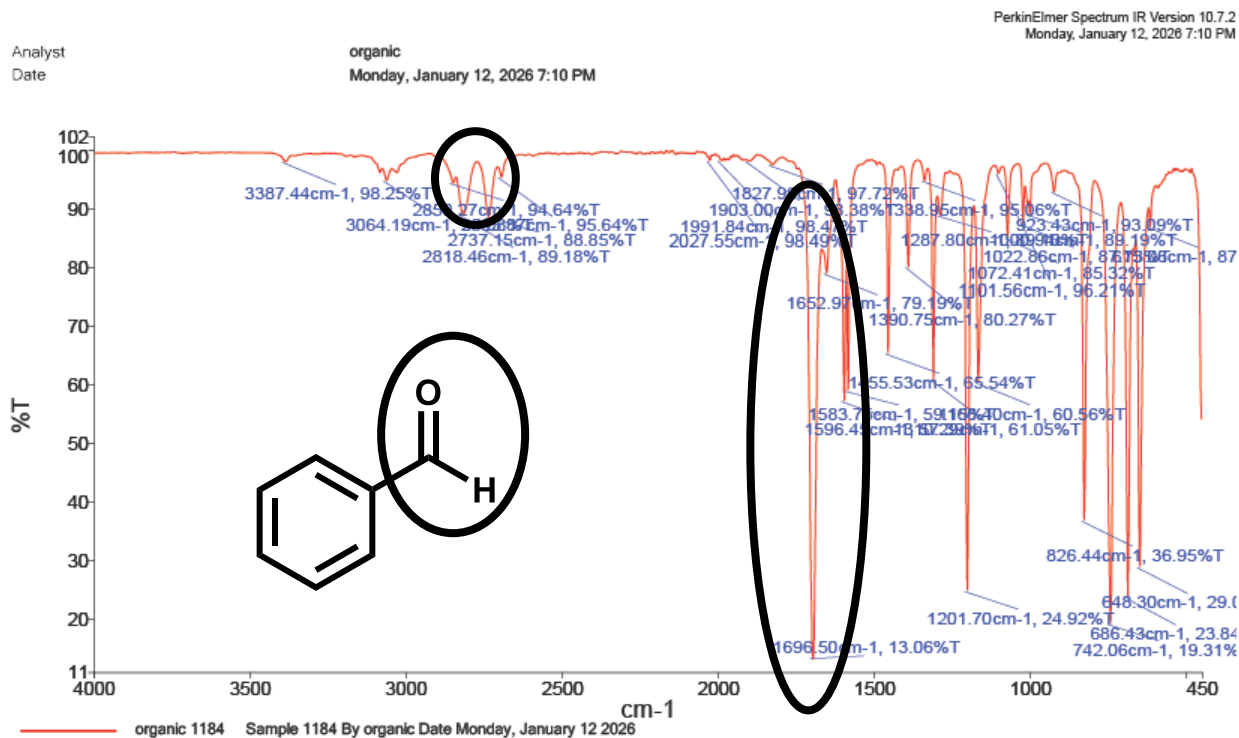
Expectations:

In-Class (0 points):

Expectations

- Must have: table filled out for hexane, hexene, 1-heptyne, heptylcyanide, toluene, hexanol, hexylamine, heptaldehyde, 3-heptanone, and heptanoic acid including:
 - Chemical structure
 - Functional groups present in molecule
 - Bonds present in functional groups
 - Theoretical stretch
 - Calculated stretch (WebMO data)
- -1 point for each row containing a missing or incorrect value

Molecule	Functional Groups	Bonds	Theoretical Stretch (cm ⁻¹)	Calculated Stretch (cm ⁻¹)
	alkyl	sp ³ C-H	3100-2850	3036, 3109
	alkenyl	sp ² C-H, C=C	3100-2850	3107, 3034, 1530
	alkynyl	sp C-H, C≡C	3300, 2260-2100	3500, 2246
	nitrile	C≡N	2260-2100	2369
	aromatic/alkyl	C=C, sp ³ C-H, sp ² C-H	1600, 3100-2850, 3100-2850	1673, 3046, 3107, 3200
	alcohol	O-H	3400-3200	3754
	amine	N-H	3500-3300	3546, 3457
	aldehyde	aldehyde C-H, C=O	2850 and 2750, 1800-1650	2888, 1835
	ketone	C=O	1800-1650	1812
	carboxylic acid	O-H, C=O	3300-2500, 1800-1650	3724, 1883



Page 1

annotation	functional group	bond and vibration	wavenumber (cm ⁻¹)
circle	aldehyde	sp ² C–H stretch	2818 and 2737
circle	aldehyde	C=O stretch	1697

Any additional analysis, such as a two-sentence explanation of what the IR data tells you about the outcome of your chemical transformation, which you will be asked for whenever possible in future labs, should be written here, directly below the IR spectrum and its associated table.

-1 point if any value in row is missing or incorrect

Out of Class (20 points)

Expectations:

An analysis of six molecules (the following bullet points are from the [Instrumental Analysis Checklist](#))

- Provide the name and draw the structure of the analyzed compound directly on the spectrum.
- Clearly annotate all relevant functional groups on the structure, then annotate the corresponding peaks on the spectrum with the same marking to illustrate the correlation.
- Create a table with the column headers shown in the example(s) below.
- Fill in the table with the annotation, corresponding functional group, any bends or stretches associated with that functional group, and the experimental wavenumber corresponding to those molecular movements.

You need to comment on any value a student is missing/reports incorrectly in a comment on Gradescope

Each row is worth 1 point

-1 point if any row contains an incorrect value

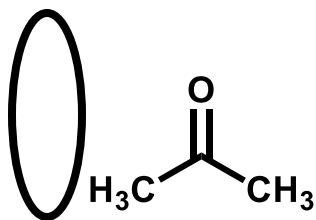
Score cannot go negative

Students may circle either the bond(s) within the functional group or the entire functional group. Both should be accepted as correct responses.

For each compound:

-1 point if missing structure with circled/labeled functional group(s) or bond(s)

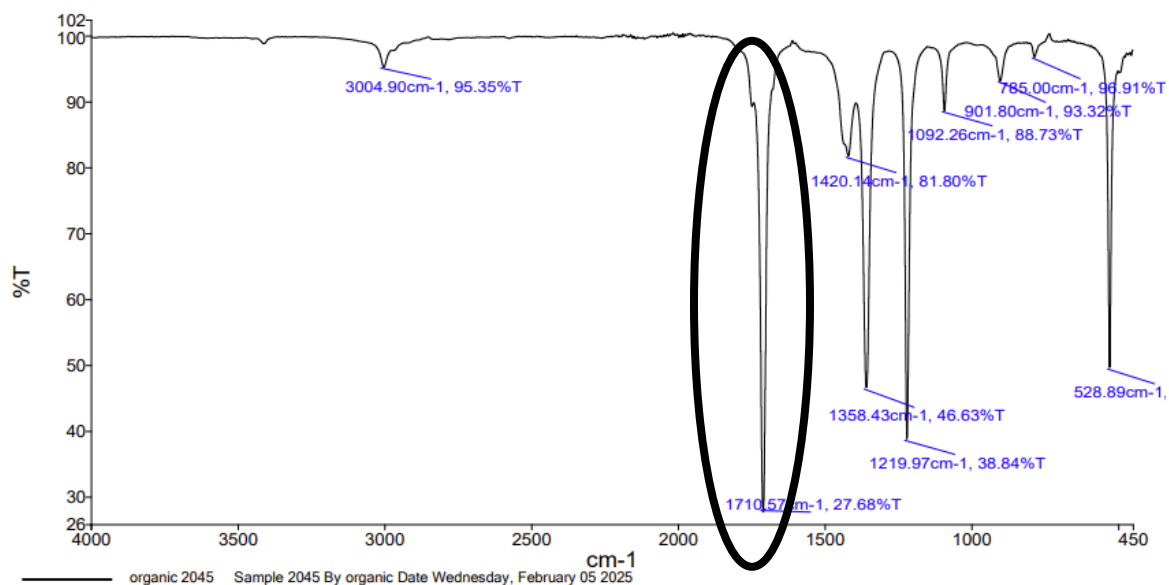
-1 point if student did not circle or label or otherwise correlate relevant stretch(es) on spectrum



PerkinElmer Spectrum IR Version 10.7.2
Wednesday, February 5, 2025 4:36 PM

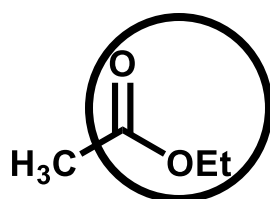
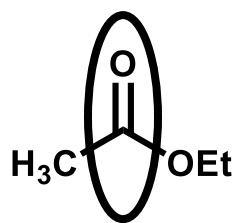
Analyst
Date

organic
Wednesday, February 5, 2025 4:36 PM

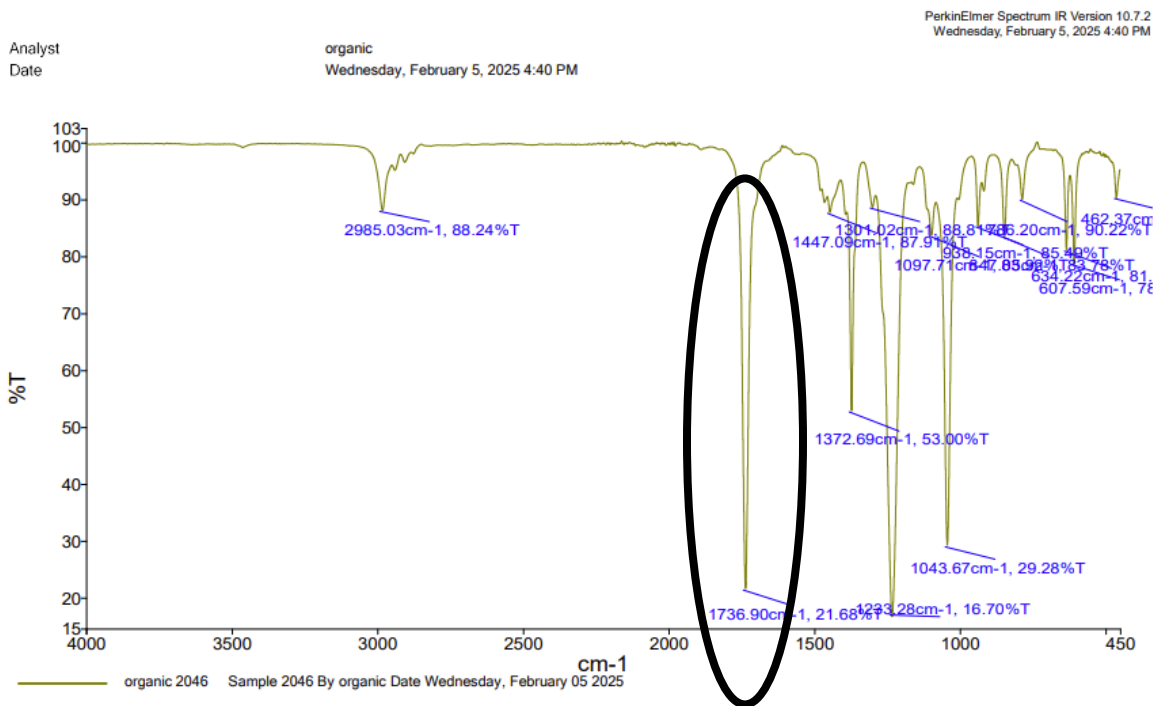


annotation	functional group	bond and vibration	Wavenumber (cm ⁻¹)
circle	ketone	C=O stretch	1710

-1 point if any value in row is incorrect

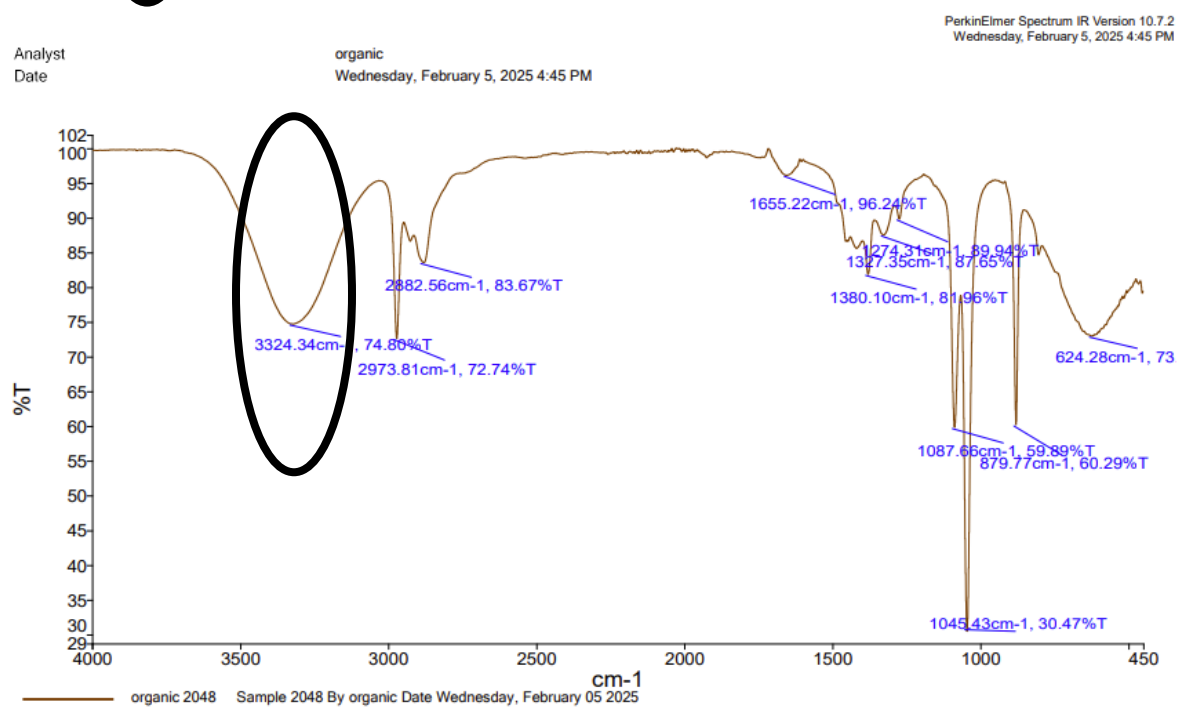
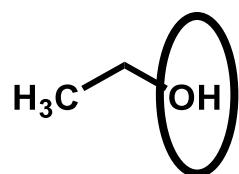


Either method of circling is correct



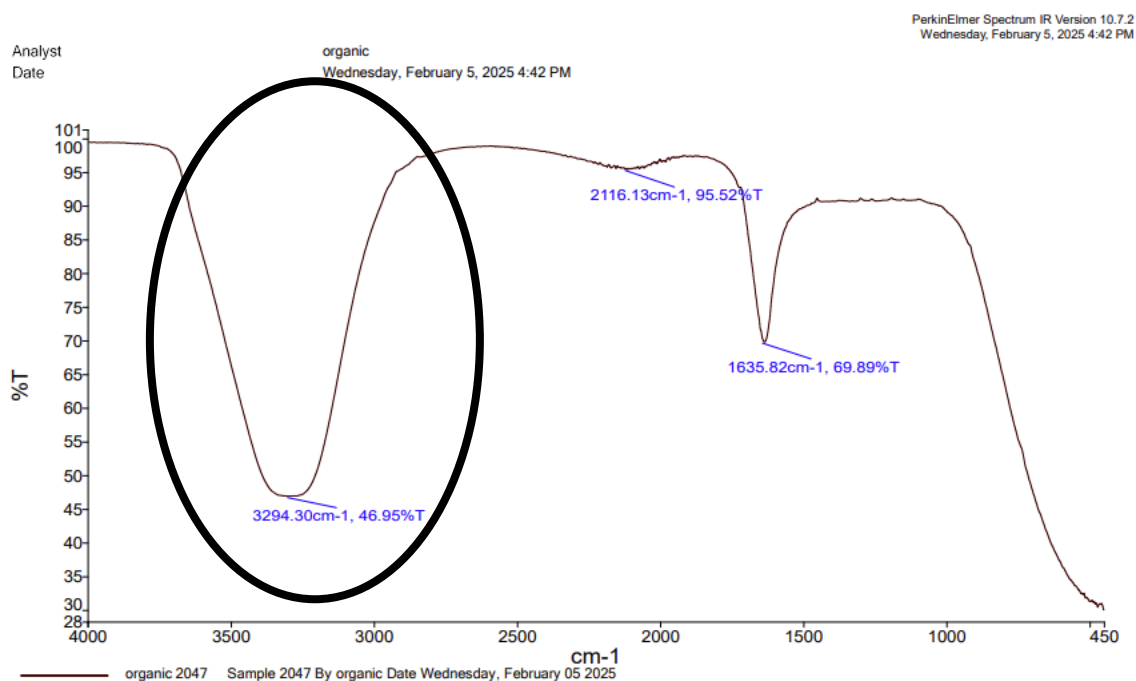
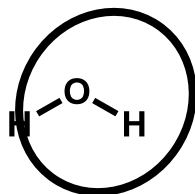
annotation	functional group	bond and vibration	Wavenumber (cm ⁻¹)
circle	ester	C=O stretch	1725

-1 point if any value in row is incorrect



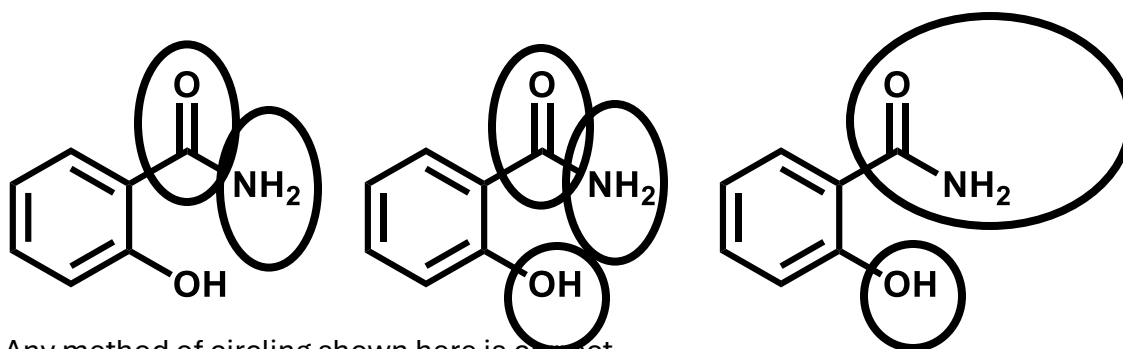
annotation	functional group	bond and vibration	Wavenumber (cm ⁻¹)
circle	alcohol	O-H stretch	3324

-1 point if any value in row is incorrect



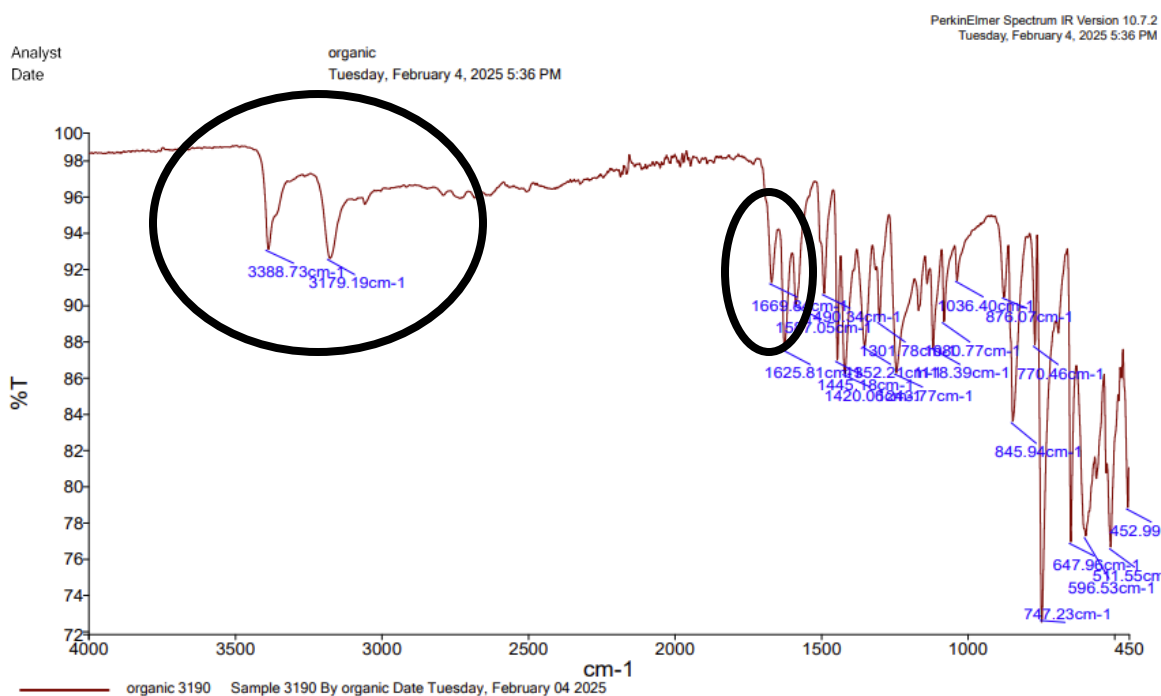
annotation	functional group	bond and vibration	Wavenumber (cm ⁻¹)
circle	hydroxyl group	O-H stretch	3294

-1 point if any value in row is incorrect



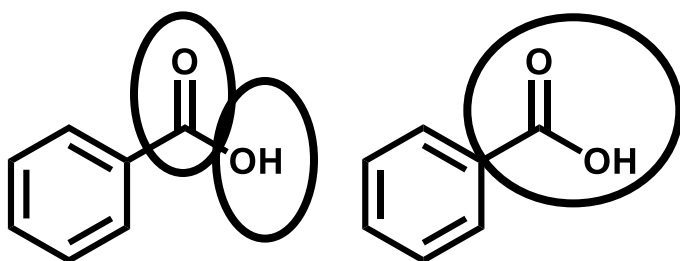
Any method of circling shown here is correct

Students may add a row to the table for the phenolic O-H stretch (around 3200 cm^{-1}), which could be so broad it is hidden under the N-H stretches. Students do not need to include a circle or row (for a total of 5 rows in the table) for this information, but they should not have points deducted if they include it.

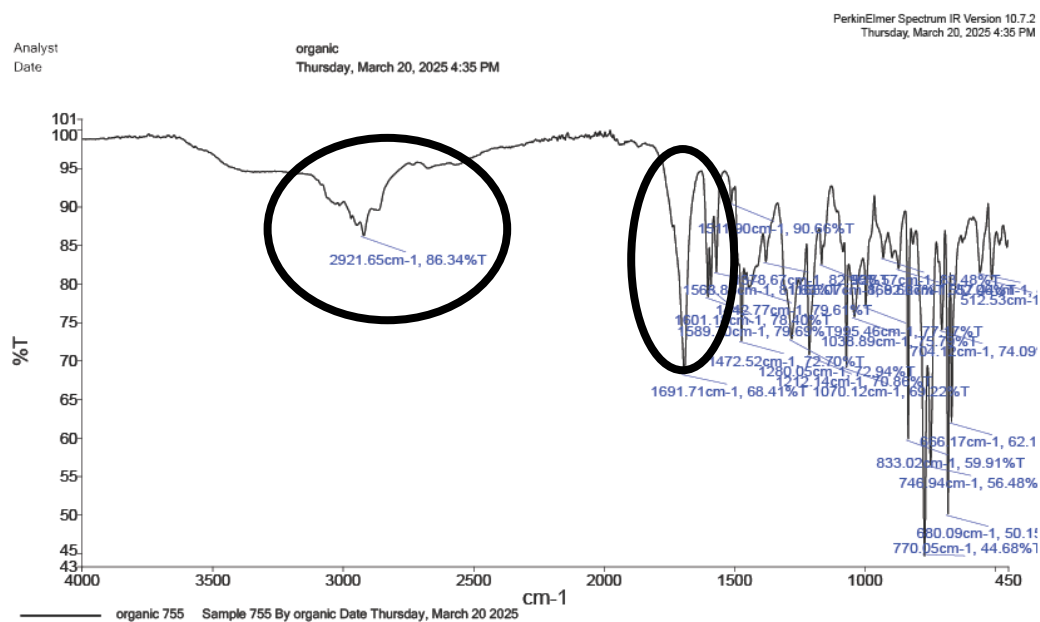


annotation	functional group	bond and vibration	Wavenumber (cm ⁻¹)
circle	amide	N-H stretch (asymmetric)	3388
circle	amide	N-H stretch (symmetric)	3179
circle	amide	C=O stretch	1670

-1 point for each row containing any incorrect value



Any method of circling shown here is correct



Page 1

annotation	functional group	bond and vibration	Wavenumber (cm ⁻¹)
circle	carboxylic acid	O-H stretch	2922
circle	carboxylic acid	C=O stretch	1682

-1 point for each row containing any incorrect value